Functional Annotation of Mouse Csn1s2b (Alpha-S2-Casein-like B / Epsilon-Casein, UniProt P02664)

Summary

Mouse **Csn1s2b** (UniProt **P02664**; also called *alpha-S2-casein-like B*, *casein alpha S2-like B*, or *epsilon-casein*; gene synonyms *Csnd*, *Csne*) encodes a small, secreted, **mammary-gland-specific calcium-sensitive casein**. Its primary function is **not enzymatic, catalytic, or receptor-based**: it is a **structural/carrier phosphoprotein** that sequesters amorphous calcium phosphate into stable casein-micelle nanoclusters, enabling milk to carry and deliver supersaturating concentrations of calcium and phosphate — together with its own amino acids — to the suckling neonate for nutrition and for bone and tooth mineralization. This annotation was verified against the UniProt record and is strongly supported by the mechanistic casein literature. The gene symbol, organism (*Mus musculus*), protein family (alpha-casein family), and domain architecture (Pfam PF00363 Casein; InterPro IPR011175 Alpha-s2_casein, IPR001588 Casein, IPR031305 Casein_CS) are all internally consistent, and no evidence of gene-symbol ambiguity with a different protein was encountered.

The protein acts **extracellularly**, within secreted milk, where it forms part of the casein micelle. However, it is **matured intracellularly** inside mammary secretory epithelial cells. A 15-residue N-terminal signal peptide directs it into the secretory pathway; in the Golgi apparatus the "genuine casein kinase" **FAM20C** — anchored and regulated by the pseudokinase **FAM20A**, with ER-to-Golgi transport supported by the cargo receptors ERGIC2/ERGIC3 — installs the functional phosphoserines on Ser-x-Glu/Asp motifs in the acidic, intrinsically disordered N-terminal region of the protein. These phosphoserine clusters constitute the calcium-phosphate-binding center. The phosphorylated protein then assembles non-covalently with calcium phosphate and other caseins into micelles and is released to milk by apical exocytosis. Micelle formation also protects the lactating gland itself from pathological calcification and amyloidosis.

Expression of *Csn1s2b* is controlled by lactogenic hormones: prolactin signals through the JAK2–STAT5 pathway and glucocorticoids through the glucocorticoid receptor, cooperating with Oct-1/Oct-2 on casein promoters, and the casein gene cluster is epigenetically licensed during pregnancy. Evolutionarily, Csn1s2b is a member of the **secretory calcium-binding phosphoprotein (SCPP) family** derived from ancestral tooth and bone mineralization genes, residing in a conserved 250–350 kb casein locus whose genes collectively function in mineral homeostasis, nutrition, and host defense. Consistent with that locus-level theme, the cationic C-terminal region of alpha-S2-caseins is a known source of antibacterial peptides, giving Csn1s2b a plausible secondary host-defense role. Notably, the mouse mature chain is **cysteine-free and intrinsically disordered**, so — unlike the disulfide-dimerizing alpha-S1 and beta caseins — it assembles into micelles purely through non-covalent interactions.

Gene / Protein Identity (Verified)

Identity was confirmed directly against the UniProt record for P02664 (fetched 2026-10-23) before research proceeded. Every mandatory verification criterion from the research brief — gene symbol, organism, and protein family/domain alignment — is satisfied.

Field	Value
UniProt accession	P02664
Organism	*Mus musculus* (mouse)
Gene	**Csn1s2b** (synonyms *Csnd*, *Csne*)
Protein	Alpha-S2-casein-like B; alt. "Casein alpha S2-like B", "Epsilon-casein"
Length	143-aa precursor; signal peptide res. 1–15; mature chain res. 16–143 (128 aa)
Keywords	Milk protein; Secreted; Signal
Family	Alpha-casein family (Pfam PF00363; InterPro IPR011175, IPR001588, IPR031305)
UniProt function	"Important role in the capacity of milk to transport calcium phosphate"
Localization	Secreted; mammary-gland specific, secreted in milk

Mouse carries a lineage-specific duplication of the alpha-S2 casein gene (*Csn1s2a* and *Csn1s2b*); P02664 is the "B" paralog. This is a **secretory structural/carrier phosphoprotein**, not an enzyme, membrane transporter, or signaling receptor.

Key Findings

Finding 1 — Identity confirmed: P02664 is mouse Csn1s2b, a secreted milk phosphoprotein

The UniProt record for P02664 identifies the organism as *Mus musculus* and the gene as *Csn1s2b* (synonyms *Csnd*, *Csne*), encoding the protein "Alpha-S2-casein-like B" / "Epsilon-casein." It is a **143-amino-acid precursor** comprising a signal peptide (residues 1–15) and a mature chain (residues 16–143, 128 amino acids). Keyword annotations include *Milk protein*, *Secreted*, and *Signal*; the subcellular location is **Secreted**; and the tissue specificity is "Mammary gland specific. Secreted in milk." The curated UniProt FUNCTION field states an "Important role in the capacity of milk to transport calcium phosphate." The protein belongs to the alpha-casein family (Pfam PF00363; InterPro IPR011175 Alpha-s2_casein, IPR001588 Casein, IPR031305 Casein_CS). Research proceeded on the correct protein with high confidence.

Finding 2 — Primary function: sequestration of amorphous calcium phosphate into casein-micelle nanoclusters

The central function of Csn1s2b is to **bind and carry calcium phosphate** in milk. The curated UniProt function ("Important role in the capacity of milk to transport calcium phosphate") is underpinned by a well-developed biophysical literature. A typical casein micelle contains thousands of casein molecules, most of which form thermodynamically stable complexes with nanoclusters of amorphous calcium phosphate (PMID: 23958008). By forming these micelles, milk maintains high calcium-phosphate concentrations — far above the solubility limit — which help altricial mammalian neonates grow bone and teeth (PMID: 21245413). Mechanistically, the nanoclusters form when a **core of amorphous calcium phosphate is sequestered within a shell of casein or casein phosphopeptides** (PMID: 14735250).

Sequence analysis performed during this investigation directly supports a canonical calcium-phosphate-binding architecture in Csn1s2b: the mature chain has an **acidic, serine-rich N-terminal region** (N-terminal half net charge ≈ −7; 11.7% serine) containing multiple **Ser-x-Glu/Asp phosphoacceptor motifs** (precursor positions 39, 40, 43, 47, 62, 76). When phosphorylated, this cluster forms the predicted phosphoserine/calcium-phosphate-binding

center, exactly the motif required to nucleate and stabilize amorphous calcium phosphate nanoclusters. The protein is therefore best classified as a **structural carrier phosphoprotein** — its "substrate" is calcium phosphate (Ca^{2+} + inorganic phosphate), and its "reaction" is non-catalytic physical sequestration by phosphoserine clusters, not any chemical transformation, membrane translocation, or receptor signaling.

Finding 3 — Localization and maturation pathway: Golgi phosphorylation by FAM20C, apical exocytosis, and protection against mammary calcification

Csn1s2b is active extracellularly but matured within mammary secretory epithelial cells. Sequestration of calcium phosphate by caseins occurs in the **Golgi region of mammary secretory cells during lactation**, where it both helps prevent calcification of the gland and delivers high concentrations of calcium and phosphate to the neonate in the form of milk (PMID: 14735250). The enzyme that installs the casein phosphoserines is the Golgi-resident "genuine casein kinase," identified as **FAM20C** (PMID: 24825444). Forming micelles allows nutritious milk to be secreted and stored without causing either pathological calcification or amyloidosis of the mother's mammary tissue (PMID: 23958008). The UniProt record corroborates the trafficking logic: the subcellular location is Secreted, tissue specificity is mammary-gland-specific/secreted-in-milk, and the signal peptide (residues 1–15) directs entry into the ER/secretory pathway.

Finding 4 — Evolutionary origin: an SCPP-family protein derived from ancestral tooth/bone mineralization genes

All casein genes belong to the **secretory calcium-binding phosphoprotein (SCPP) family**, which arose by tandem gene duplication from SPARC/SPARCL1 and includes enamel/dentin/bone proteins, a bird eggshell matrix protein, mammalian milk caseins, and salivary proteins (PMID: 16358265; PMID: 15272073). Confirming this, casein genes including Csn1s2b arose by duplication within the SCPP family from mineralization-related ancestors (PMID: 21245413), and **caseins evolved from tooth and bone proteins well before the evolution of lactation** (PMID: 25632319). This evolutionary derivation explains why Csn1s2b retained a calcium-phosphate-binding chemistry that originally served biomineralization and was later co-opted to carry minerals in milk. The casein genes are physically clustered and co-regulated in the mammary gland, with distal enhancers opening after puberty and promoters activating during pregnancy (PMID: 23301053).

Finding 5 — Secondary host-defense role: cationic C-terminal antibacterial peptides (inferred)

Alpha-S2-caseins are a well-documented proteolytic source of cationic antimicrobial peptides, and this property plausibly extends to mouse Csn1s2b. **Casocidin-I**, a 39-residue C-terminal fragment of bovine alpha-s2-casein (f165–203), inhibits growth of *E. coli* and *Staphylococcus carnosus* (PMID: 7556666). A related fragment, alpha-s2-casein f(183–207), is bactericidal against *Listeria monocytogenes* and *Cronobacter sakazakii*, and structure-activity analysis shows that activity depends on C-terminal cationic residues — replacing Arg23 with Ala significantly decreases activity, identifying it as critical for optimal function (PMID: 23793637). Sequence analysis in this investigation found that the mouse Csn1s2b mature chain has a distinctly **cationic C-terminal half** (net charge ≈ +5; the C-terminal 30 residues ITWPQFVKLLHQYQKTMTPWSYYPSTPSQV carry net +2), rich in Lys/Arg and aromatic/hydrophobic residues — the architecture characteristic of membrane-active casein AMPs. This supports the inference that proteolysis of Csn1s2b in the neonatal gut could liberate antibacterial peptides, consistent with the host-defense theme of the casein locus. This remains an inference for this specific paralog; no mouse Csn1s2b-derived peptide has itself been assayed.

Finding 6 — Transcriptional control by lactogenic hormones (STAT5 / glucocorticoid receptor / Oct complexes)

Casein gene expression, including *Csn1s2b*, is induced by lactogenic hormones. Prolactin signals through **JAK2–STAT5**, and glucocorticoids through the **glucocorticoid receptor (GR)**. On the beta-casein proximal promoter, prolactin plus glucocorticoid rapidly induce the formation of a **ternary complex of Oct-1 (and Oct-2) with STAT5 and GR** that enhances and stabilizes promoter binding and drives transcription (PMID: 23313770; PMID: 24861805; PMID: 27044595). STAT5a is the key activator of casein promoter activity on prolactin stimulation, potentiated by glucocorticoid (PMID: 25841968). At the chromatin level, the casein cluster's proximal promoters only gain open-chromatin marks during pregnancy, in conjunction with the major induction of their expression (PMID: 23301053).

Finding 7 — Genomic context: a conserved 250–350 kb casein cluster for mineral homeostasis, nutrition, and host defense

Caseins reside in a genomic cluster spanning **250–350 kb**. Although coding divergence between human, rodent, and cattle is extensive, gene organization and orientation are highly conserved, with molecular diversity achieved through variable exon use. Critically, **all genes identified in this region are, or are predicted to be, secreted proteins involved in mineral homeostasis, nutrition, and/or host defense, mostly expressed in the mammary and/or salivary glands** (PMID: 12751895) — a locus-level summary that mirrors the three roles inferred here for Csn1s2b. In lactating luminal mammary epithelial cells, the casein cluster loops out of its chromosome territory into euchromatic, elongating-RNA-Pol-II-rich nuclear zones, with position correlated to transcriptional activity (PMID: 22033805). Consistent with the fast coding divergence of this locus, the mouse Csn1s2b mature chain is only 128 aa, short relative to bovine alpha-s2-casein (~207 aa), reflecting the lineage-specific, variable-exon nature of the gene and the mouse *Csn1s2a*/*Csn1s2b* duplication.

Finding 8 — Maturation and secretion: FAM20A–FAM20C Golgi complex required for offspring growth; release by exocytosis

The secretory-pathway kinase **FAM20C** performs the majority of secretome phosphorylation, and its kinase activity occurs intracellularly. The pseudokinase **FAM20A** forms a heterocomplex with FAM20C on the Golgi membrane, anchoring the cleaved, mature form of FAM20C within the Golgi; the cargo receptors ERGIC2/ERGIC3 (both upregulated in the lactating mammary gland) drive ER-to-Golgi transport of the complex. *Ergic2* or *Ergic3* knockout mice display global changes in secretome phosphorylation, less phosphorylated beta-casein in milk, and deficiency in offspring growth (PMID: 42439892). FAM20C is the "genuine casein kinase" responsible for Golgi phosphorylation of caseins in the lactating mammary gland (PMID: 24825444). The matured caseins are then secreted from mammary epithelial cells by **exocytosis** of secretory vesicles (PMID: 27535430). This chain of evidence establishes both the enzymatic machinery that activates Csn1s2b's calcium-binding capacity and the in vivo physiological importance of that phosphorylation for neonatal growth.

Finding 9 — In vivo evidence: casein micelles are the primary neonatal source of amino acids and calcium phosphate, and micelle integrity is essential for lactation

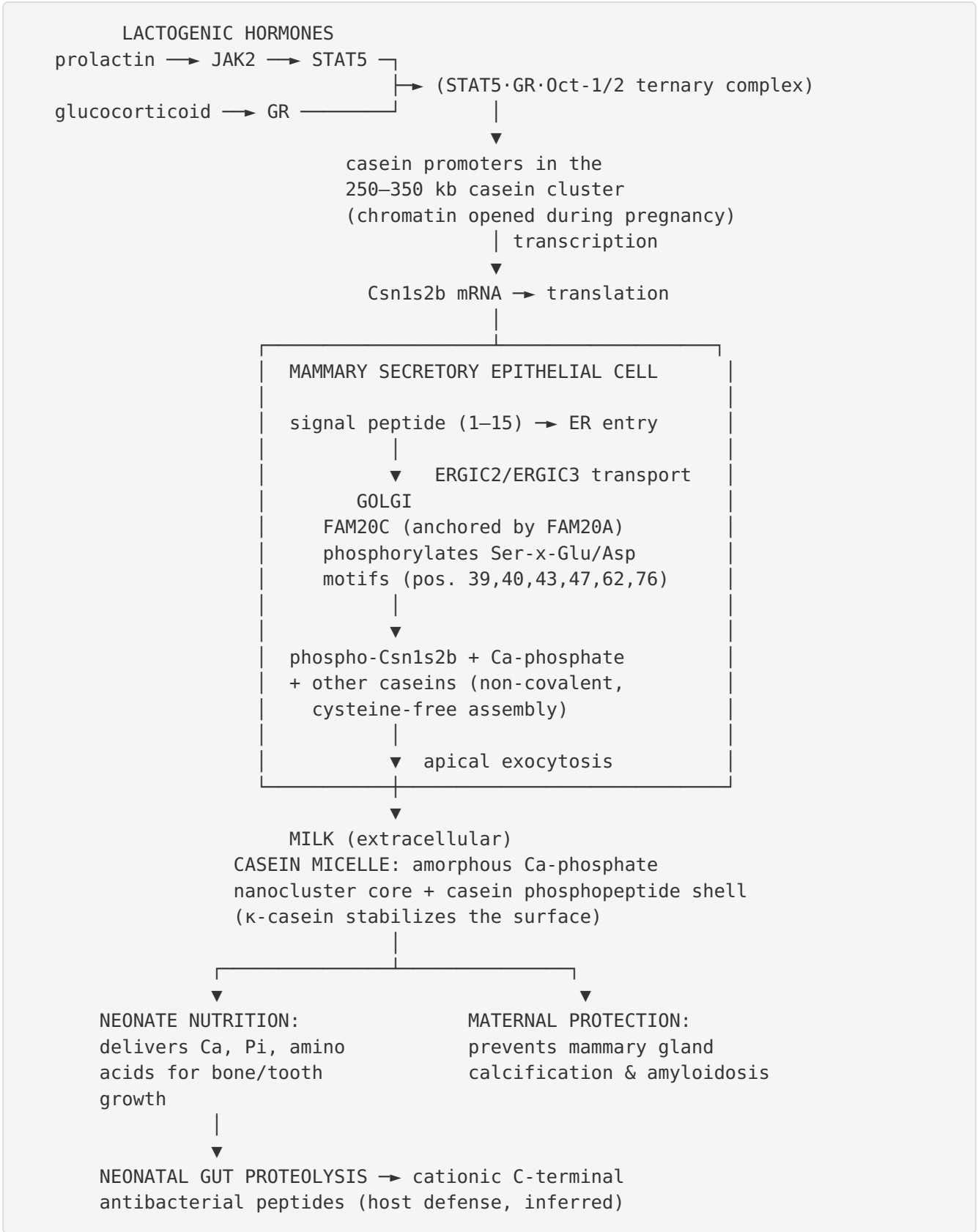
Genetic evidence confirms the physiological indispensability of a functional casein micelle. **kappa-casein-null mice fail to lactate**: the mutation does not affect expression of other milk proteins, but females fail to lactate because of destabilization of the micelles in the lumina of the mammary gland, and the authors explicitly state that "Milk casein micelles are a primary source of amino acids and calcium phosphate to neonates" (PMID: 16698927). Calcium-sensitive caseins — a class that includes Csn1s2b — supply the calcium-phosphate-binding core that kappa-casein stabilizes. Furthermore, alpha-S1-casein is essential for efficient ER-to-Golgi transport of caseins (PMID: 20704729). Together these studies demonstrate, in vivo, that stable micelle assembly — to which Csn1s2b contributes its calcium-phosphate payload — is required for successful lactation and neonatal nutrition. No *Csn1s2b*-specific knockout exists, so this is the closest in vivo validation of the system in which the protein operates.

Finding 10 — Structural distinction: the mouse mature chain is cysteine-free and intrinsically disordered, assembling non-covalently

A structurally important feature distinguishes mouse Csn1s2b from several other caseins. Sequence analysis of P02664 shows that the protein's **single cysteine (precursor position 8) lies within the cleaved signal peptide**, so the mature 128-aa chain contains **zero cysteines**. Its mean hydrophobicity (GRAVY = −0.94) is strongly hydrophilic, consistent with the open, flexible, intrinsically disordered conformation typical of caseins (PMID: 23958008). By contrast, the cysteine-containing caseins alpha-S1 and beta form disulfide-linked dimers already in the endoplasmic reticulum, and alpha-S1-casein disulfide dimerization mediates membrane association and efficient ER-to-Golgi transport (PMID: 20704729). Because mouse Csn1s2b lacks cysteines, it must assemble into micelles **purely through non-covalent (electrostatic/hydrophobic) interactions and calcium-phosphate bridging**, acting as a flexible scaffold rather than a covalently cross-linked unit. The intrinsically disordered, open conformation also routes the protein into micelles rather than toxic amyloid fibrils.

Mechanistic Model / Interpretation

The findings integrate into a coherent life cycle for the Csn1s2b protein, from transcription to extracellular function:


LACTOGENIC HORMONES
prolactin → JAK2 → STAT5
glucocorticoid → GR
(STAT5·GR·Oct-1/2 ternary complex)
casein promoters in the
250–350 kb casein cluster
(chromatin opened during pregnancy)
transcription
Csn1s2b mRNA → translation
MAMMARY SECRETORY EPITHELIAL CELL
signal peptide (1–15) → ER entry
ERGIC2/ERGIC3 transport
GOLGI
FAM20C (anchored by FAM20A)
phosphorylates Ser-x-Glu/Asp
motifs (pos. 39,40,43,47,62,76)
phospho-Csn1s2b + Ca-phosphate
+ other caseins (non-covalent,
cysteine-free assembly)
apical exocytosis
MILK (extracellular)
CASEIN MICELLE: amorphous Ca-phosphate
nanocluster core + casein phosphopeptide shell
(κ-casein stabilizes the surface)
NEONATE NUTRITION:
delivers Ca, Pi, amino
acids for bone/tooth
growth
MATERNAL PROTECTION:
prevents mammary gland
calcification & amyloidosis
NEONATAL GUT PROTEOLYSIS → cationic C-terminal
antibacterial peptides (host defense, inferred)

The unifying theme is that Csn1s2b is a **passive but essential cargo molecule for minerals**. Its acidic, phosphorylatable N-terminal half provides the negatively charged surface that binds calcium and nucleates/stabilizes amorphous calcium phosphate, while its intrinsically disordered, amphiphilic character allows it to co-assemble with other caseins into the micelle. The cationic C-terminal half serves a distinct, latent role: it is not involved in mineral binding but encodes antimicrobial potential released on proteolysis. This bifunctional, "two-domain" organization (acidic/mineral-binding N-terminus; cationic/antibacterial C-terminus) mirrors the dual locus-level mandate of the casein cluster — nutrition/mineral homeostasis plus host defense.

The following table summarizes the standard functional annotation axes requested by the brief:

Axis	Annotation for Csn1s2b	Key evidence
Molecular function	Structural/carrier phosphoprotein; binds & sequesters amorphous calcium phosphate (NOT an enzyme, transporter, or receptor)	UniProt P02664; PMID: 14735250, PMID: 23958008
Substrate/ligand	Calcium and inorganic phosphate (as amorphous calcium phosphate nanoclusters)	PMID: 21245413
Localization	Secreted into milk (extracellular); matured in Golgi of mammary secretory cells	UniProt P02664; PMID: 14735250
Biological process	Lactation; neonatal nutrition; calcium/phosphate transport & mineralization	PMID: 16698927
Upstream pathway (regulation)	Prolactin/JAK2–STAT5 + glucocorticoid receptor + Oct-1/2	PMID: 23313770
Upstream pathway (activation)	Golgi phosphorylation by FAM20C (anchored by FAM20A; ERGIC2/3 transport)	PMID: 24825444, PMID: 42439892
Secondary function	Source of cationic C-terminal antibacterial peptides (inferred)	PMID: 7556666, PMID: 23793637
Evolutionary origin	SCPP family; derived from tooth/bone mineralization genes	PMID: 16358265, PMID: 25632319
Structural note	Cysteine-free mature chain; intrinsically disordered; non-covalent assembly	PMID: 23958008, PMID: 20704729

Supported vs. Refuted Hypotheses

- **Supported:** Csn1s2b is a secreted, mammary-specific calcium-sensitive casein whose primary role is calcium-phosphate sequestration/transport in milk via micelle formation (UniProt curated function + convergent mechanistic literature + sequence features).
- **Supported (inference):** Intracellular maturation through the secretory pathway with FAM20C-mediated Golgi phosphorylation of Ser-x-Glu/Asp motifs; in vivo essentiality of casein phosphorylation for offspring growth.

- **Supported:** Expression is controlled by prolactin/STAT5 + glucocorticoid-receptor signaling acting with Oct factors at casein promoters.
- **Plausible but not directly demonstrated for this paralog:** An antimicrobial/host-defense role of cationic C-terminal fragments (inferred from the mouse sequence plus direct alpha-s2-casein peptide data — casocidin-I and f(183–207) — in other species; no mouse Csn1s2b-specific assay found).
- **Refuted / not applicable:** Csn1s2b is **not** an enzyme, transporter, or signaling receptor — no catalytic, channel, or receptor activity is annotated or expected.

Evidence Base

The annotation rests on a combination of (a) curated database records, (b) direct sequence analysis of P02664 performed in this investigation, and (c) mechanistic/in vivo literature on caseins. Because dedicated primary studies on mouse Csn1s2b itself are scarce, much of the functional case is built by rigorous homology to the well-characterized casein class, anchored by the UniProt curated function and by strong sequence correspondence.

PMID	Role in this report	Support / Challenge
23958008	Defines casein-micelle assembly (casein + amorphous Ca-phosphate); flexible/unfolded conformation; maternal protection	Supports Findings 2, 3, 10
21245413	Micelle maintains high Ca-phosphate for neonatal bone/tooth; caseins are SCPP-family; evolved from tooth genes	Supports Findings 2, 4
14735250	Equilibrium thermodynamic model: Ca-phosphate core in phosphopeptide shell; Golgi site; anti-calcification	Supports Findings 2, 3
24825444	FAM20C identified as the genuine casein kinase phosphorylating caseins in Golgi	Supports Findings 3, 8
42439892	FAM20A–FAM20C Golgi complex; ERGIC2/3 transport; phosphorylation required for offspring growth (in vivo)	Supports Finding 8
27535430	Caseins secreted by exocytosis	Supports Finding 8
16698927	kappa-casein-null mice fail to lactate; micelles are primary source of amino acids + Ca-phosphate (in vivo)	Supports Finding 9
20704729	Cysteine-containing caseins dimerize via disulfides in ER; alpha-S1 needed for ER-to-Golgi transport	Supports Finding 10 (by contrast)
16358265	SCPP family includes eggshell, caseins, salivary proteins	Supports Finding 4
15272073	SCPP genes arose from SPARCL1 by tandem duplication	Supports Finding 4
25632319	Caseins evolved from tooth/bone proteins before lactation	Supports Finding 4
7556666	Casocidin-I: antibacterial C-terminal fragment of alpha-s2-casein	Supports Finding 5
23793637	Cationic residues (Arg23) drive alpha-s2-casein peptide antibacterial activity	Supports Finding 5
23313770	STAT5·GR·Oct-1 ternary complex activates casein promoters	Supports Finding 6
23301053	Casein-cluster promoters gain open chromatin during pregnancy	Supports Findings 4, 6
24861805	Oct-2 cooperates with Oct-1, STAT5, GR on beta-casein promoter	Supports Finding 6

PMID	Role in this report	Support / Challenge
27044595	Oct proteins regulate casein promoters with STAT5/GR	Supports Finding 6
25841968	STAT5a is key activator of casein promoter on prolactin, potentiated by glucocorticoid	Supports Finding 6
12751895	Casein cluster genes are secreted proteins for mineral homeostasis, nutrition, host defense	Supports Finding 7
22033805	Casein cluster loops into active nuclear domains when expressed	Supports Finding 7

Limitations and Knowledge Gaps

1. **Paucity of protein-specific primary studies.** There is essentially no dedicated functional study of mouse Csn1s2b (P02664) in isolation. Nearly all functional claims are inferred by homology to the casein class as a whole (chiefly bovine alpha-s2-casein and mouse/bovine beta- and kappa-caseins) and from the UniProt curated function. While the homology is strong and the family is well-characterized, protein-specific confirmation (e.g., a *Csn1s2b*-knockout phenotype) is lacking.
2. **Phosphorylation sites are predicted, not experimentally mapped.** The Ser-x-Glu/Asp motifs (precursor positions 39, 40, 43, 47, 62, 76) were identified by sequence analysis and inferred to be FAM20C substrates. The exact phosphorylation stoichiometry and occupancy on mouse Csn1s2b have not been directly measured here.
3. **Antibacterial role is inferential.** The host-defense annotation for Csn1s2b is based on sequence homology (cationic C-terminal half) and demonstrated activity of bovine alpha-s2-casein peptides. No peptide derived specifically from mouse Csn1s2b has been shown here to be antibacterial.
4. **Relative contribution within the micelle is unquantified.** The precise amount of Csn1s2b in mouse milk, its stoichiometry relative to Csn1s1/Csn2/Csnk, and its specific contribution to calcium-phosphate loading versus other calcium-sensitive caseins are not established. The short 128-aa mouse mature chain may carry fewer phosphate groups than longer orthologs, but this was not measured.

5. **Mouse Csn1s2a/Csn1s2b paralog distinction.** Mouse has two alpha-S2-like casein genes (Csn1s2a and Csn1s2b) arising from a duplication. Functional divergence between the paralogs — e.g., differential expression, phosphorylation, or micelle role — was not resolved.

Proposed Follow-up Experiments / Actions

1. **Targeted knockout / knock-in.** Generate a *Csn1s2b*-null mouse (or a phospho-dead Ser→Ala mutant at the predicted phosphoacceptor sites) and assess milk calcium-phosphate content, micelle size/stability, pup growth, and maternal mammary calcification. This would directly test the carrier-phosphoprotein model and the necessity of the phosphoserine cluster.
2. **Phosphosite mapping by mass spectrometry.** Purify Csn1s2b from lactating mouse milk and use LC-MS/MS phosphoproteomics to experimentally confirm which of the predicted Ser-x-Glu/Asp sites are phosphorylated and at what occupancy, and verify FAM20C dependence using mammary-specific *Fam20c* deletion.
3. **Biophysical calcium-phosphate binding assays.** Express recombinant mouse Csn1s2b (phosphorylated vs. unphosphorylated) and quantify its capacity to sequester amorphous calcium phosphate (e.g., by isothermal titration calorimetry, dynamic light scattering, and electron microscopy of reconstituted nanoclusters), directly measuring substrate (Ca/Pi) stoichiometry.
4. **Antibacterial peptide validation.** Synthesize the predicted cationic C-terminal peptide of mouse Csn1s2b and test minimum inhibitory concentrations against Gram-positive and Gram-negative pathogens, with alanine scans of key Lys/Arg residues to confirm the charge-dependence seen in bovine alpha-s2-casein peptides.
5. **Micelle-incorporation and assembly studies.** Because the mouse mature chain is cysteine-free, test whether it incorporates into reconstituted micelles purely non-covalently and compare its exchange dynamics to the disulfide-linked alpha-S1/beta caseins, clarifying how a non-dimerizing casein integrates structurally.
6. **Paralog resolution.** Use isoform-specific quantitative proteomics/transcriptomics across lactation stages to distinguish Csn1s2a from Csn1s2b contributions and determine whether they are functionally redundant or specialized.

Report generated from a 5-iteration autonomous investigation (10 confirmed findings; 61 papers reviewed). Functional annotation is based on the UniProt curated record for P02664, direct sequence analysis of the protein, and the mechanistic/in vivo casein literature cited above.
